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Université de Lausanne
Faculté des Hautes Études Commerciales

Quantitative Asset Management II

Quality at Good Price

Abstract

This paper presents a comprehensive analysis of systematic long/short equity trading strategies, emphasizing the integration of quality into traditional value investing for the US equity market. By examining 5760 portfolios, through an hyperparameterization process across five long-short exposure profiles, we demonstrate that a combined long value-quality and short (poor) quality approach significantly outperforms a value-only strategy in terms of risk-adjusted performance. Our sensitivity analysis, leveraging extensive data from various sources, validates the robustness of this pattern across diverse market settings. The paper highlights the importance of tailoring stock selection to client risk profiles, showcasing the potential for superior risk-adjusted returns with an integrated value-quality strategy.

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1 Introduction

In this project, we present a meticulously designed strategy tailored for our client, a seasoned discretionary hedge fund manager. With a profound admiration for the investment philosophy of Warren Buffett, our client embraces the principle of not overpaying for stocks. This approach has served as the cornerstone of his investment decisions. However, the recent challenges faced by the value style have prompted a significant shift in his perspective, motivating him to explore the integration of "quality" as a critical factor in his investment strategy.

Our objective is to develop a systematic strategy that aligns with the concept of "quality at a good price", specifically targeting the US equity market. The strategy is to be highly selective, with the hedge fund's portfolio comprising no more than 25 stocks each on the long and short sides, thereby allowing for concentrated positions. Furthermore, a key requirement from our client is a comprehensive analysis of the performance of both the long and short legs. This analysis is not just to understand the individual performances but also to ascertain the enhanced value brought about by integrating quality into a traditional value portfolio. This report outlines the development, implementation, and potential impact of this strategy.

2 Literature Review

In the evolving landscape of investment strategies, quality investing emerges as a multi-dimension and increasingly popular approach. This strategy pivots on the identification of companies exhibiting robust financial health and stability, marked by high profitability, strong accounting quality, and sound capital structure. The essence of quality investing is captured through a comprehensive analysis of its various dimensions in recent academia papers, which collectively offer a rich understanding of its mechanics and outcomes.

Central to the exploration of quality investing is the legendary investor Warren Buffett, whose methodology has been the subject of extensive analysis. Frazzini et al., 2018's study, "Buffett's Alpha", deeps into the complexities of Buffett's investment strategy at Berkshire Hathaway. They reveal that his remarkable success, characterized by a notable Sharpe ratio and out-performance of traditional risk factors, can be largely attributed to his strategic emphasis on leveraging safe, high-quality stocks. The study uncovers that Buffett's alpha becomes less significant when adjusting for his exposure to specific factors like "betting against beta" and "quality minus junk". This analysis not only highlights Buffett's exceptional stock-picking acumen but also his effective use of leverage, primarily facilitated through low-cost insurance float, which amplifies his returns.

Complementing these insights, Hsu et al., 2019, "What Is Quality", provides a detailed exploration of the "quality" factor in equity investing. In a landscape where "quality" lacks a universally accepted definition, unlike more established factors such as value and size, their investigation into various characteristics associated with quality investing is particularly relevant. The authors identify key characteristics that reliably predict future returns, including profitability, accounting quality, and payout/dilution. Employing a robust three-step procedure, they test these factors across different markets and definitions, reinforcing the empirical basis for quality investing.

Further enriching the understanding of quality investing, Asness, Frazzini, and Pedersen, 2019's study "Quality Minus Junk" examines how quality characteristics in stocks affect their performance and pricing. They define quality in terms of traits that investors should pay a higher price for, such as profitability, growth, and safety. Their research uncovers that high-quality stocks command higher prices but not by a significantly large margin, leading to high risk-adjusted returns for these stocks. The introduction of the "quality-minus-junk" (QMJ) factor, which involves long positions in high-quality stocks and shorts in low-quality stocks, demonstrates significant risk-adjusted returns in various markets globally. This factor's resilience even during market downturns challenges conventional risk-based explanations and underscores the robustness of quality stocks.

Our project aims to apply these empirical findings to develop a comprehensive long/short strategy for our client. Drawing from the methodologies and insights of these studies, we plan to construct a portfolio that not only identifies high-quality stocks for long positions but also discerns potential short opportunities in overvalued or financially weaker stocks. The goal is to harness the proven advantages of quality investing, by integrating quality together with value and assess the benefits of this integration. Indeed, integrating investing styles has been shown already by Fitzgibbons et al., 2017 to improve excess returns, however in a long-only setting.

3 Data Analysis

We exploit datasets sourced from three data providers, offering a comprehensive view of US equities traded on three major stock exchanges, namely NYSE, NASDAQ, and the former American Stock Exchange (AMEX).

Leveraging the fundamentals quarterly data retrieved from CRSP Compustat Merged, we undertake a detailed examination of balance sheet and income statement metrics for numerous US firms. This dataset covers financial indicators gauging earnings, assets, liabilities, and operating incomes.

In parallel, the CRSP's security monthly data offers a detailed perspective on total stock returns, calculated on a monthly basis. Additionally, the stock monthly data from CRSP contributes information concerning trading volumes and returns of the S&P 500 Composite Index, which is used as market proxy. Complementing these datasets, the incorporation of the Fama-French 3-factors plus momentum (FF3-MoM) data from the WRDS Fama French database broadens the analytical framework. Notably, throughout the study, we use as risk-free rate the 1-month US Treasuries yield, which serves as a benchmark for evaluating risk-adjusted returns.

3.1 Data Management

Our data preprocessing entails several steps to ensure the integrity and coherence of our final dataset. First, we identify row duplicates across multiple datasets, thus removing such entries to maintain integrity. Subsequently, we create the necessary keys to merge diverse datasets into a singular, cohesive dataset.

Recognizing the significance of temporal completeness, particularly concerning delisting and relisting events impacting time continuity, we methodically address missing dates. Absent dates during stock transition periods are reconciled by infusing missing values and corresponding keys, ensuring a seamless and complete chronological continuum within each stock's existence period. This meticulous calendar normalization fortifies our dataset's temporal integrity, allowing for subsequent processes.

Furthermore, our preprocessing phase include the creation of additional accounting metrics derived from available fundamental data, pivotal for the score building process. Also, a crucial step towards ensuring the out-of-sample nature of our analysis involves a forward shift by one quarter of all fundamental data. This adjustment aligns our dataset with the practical investment decision-making timeline, where data published at the end of a quarter becomes available for investment considerations only in the subsequent quarter's initial month. This forward push enhances the authenticity and real-world applicability of our findings.

3.2 Data Description

Our final dataset spans from December 1994 to December 2022, encompassing stocks across all 11 GICS sectors. The stocks universe, initially comprising 358 entities in 1994 and expanding to 1427 entities by 2022, is illustrated in Figure 1, which underscores the consistent presence of all sectors across the majority of the observed period.

In order to prevent potential liquidity issues, we device two hard filters to refine the dataset. First, we exclude companies exhibiting minimal monthly trading volume, specifically those with trading volumes lower than $25 mil. Additionally, we filter out small companies with a market capitalization inferior to $250 mil. These filters enhance the dataset's robustness, ensuring a focus on entities with significant market presence while excluding less liquid and inaccessible stocks.

In the Appendix, we provide a comprehensive table describing all the variables encompassed in our analysis (refer to Table 1).

4 Methodology

4.1 Score Building

Our analysis's principal objective is to assess if integrating the quality factor with the value factor can lead to a better-performing portfolio compared to a value-only long/short portfolio. To do that, we need to identify measures to assess which companies in our universe have the largest exposure to these two factors. In order to evaluate a specific company's exposure to these two factors, we have employed a comprehensive approach by combining multiple measures. Moreover, blending different measures to asses a factor exposure or a component exposure inside a factor, allows for more robustness. Concretely, we have computed composite scores by blending together several metrics to identify value and quality companies.

The value score comprises three distinct metrics chosen in accordance with Asness, Frazzini, Israel, et al., 2015: the book-to-market equity ratio (BE/ME), the earnings-to-price ratio (E/P), and the cash flow-to-price ratio (CF/P). Asness, Frazzini, Israel, et al., 2015 demonstrated in their study that these three ratios yield the most robust results among all value measures.

In constructing the quality score, we adhere to the methodology outlined in Asness, Frazzini, and Pedersen, 2019. This involves breaking down quality into three distinct components that measure a company's profitability, the growth in profitability, and its overall safety. Namely:

- The profitability component includes the gross profit over asset (GPOA), the return on equity (ROE), the return on asset (ROA), the cash flow over asset (CFOA), the gross margin (GMAR), and the accruals (ACC).
- The growth component considers the variation of the different profitability metrics over the last five years (excluding the accruals).
- The safety component comprises the leverage (LEV), the Altman z-score (Z), and the market beta (BAB) of the company.

One important element to mention for the accruals, leverage and, beta is that we take the negative of these measures in our score building since a lower value for these metrics results in a higher-quality company. As a result, for all our indicators, a higher value means a larger exposure to the factor the measures try to capture. The Appendix includes a table presenting precise definitions and the variables employed in calculating these measures (refer to Table 2).

After defining the various measures to assess a company's exposure to different factors, we consolidate all the variables to compute scores. For each variable used in the scores, we calculate a z-score.

At each rebalancing date t, for all the investible stocks i the z-score for a variable x is defined as:

$$z_{i,t}(x_i) = \frac{r_{i,t} - \mu_t}{\sigma_t} \quad \text{with} \quad r_i = rank(x_i) \quad \forall t$$

where μ_t and σ_t are the cross sectional mean and standard deviation.

A high z-score means that a company has a higher exposure to the factor the variable tries to measure. At each rebalancing date t, we compute the safety score by averaging the different z-scores of the value measures.

$$Value = \text{mean}(z_{BE/ME},\ z_{E/P},\ z_{CF/P})$$

We compute the quality score as the average of the profitability, growth, and safety components, for which we defined a score as the average of the z-score of all the measures inside these components.

$$\begin{aligned}
Quality &= \text{mean}(Profitability,\ Growth,\ Safety) \\
Profitability &= \text{mean}(z_{gpoa},\ z_{roe},\ z_{roa},\ z_{cfoa},\ z_{gmar},\ z_{acc}) \\
Growth &= \text{mean}(z_{\Delta gpoa},\ z_{\Delta roe},\ z_{\Delta roa},\ z_{\Delta cfoa},\ z_{\Delta gmar}) \\
Safety &= \text{mean}(z_{bab},\ z_{lev},\ z_{z})
\end{aligned}$$

Ultimately, to find value and quality companies, we derive a value and quality score by calculating the average of the value score and the quality score for each company.

$$Value\ \&\ Quality = \text{mean}(Value,\ Quality)$$

We also add a momentum factor on top of the value and quality factors. The momentum factor is defined with two components: a classical momentum factor, which measures the company stock return over the last twelve months excluding the last months (MOM), and a reversal component, which measures the return over the last month (REV). To compute the momentum score, we apply the same methodology as for the value and quality score.

$$Momentum = \text{mean}(z_{MOM},\ z_{REV})$$

Finally, we integrate value, quality, and momentum into a single score, which we will compare against value-quality.

$$Value\ \&\ Quality\ \&\ Momentum = \text{mean}(Value,\ Quality,\ Momentum)$$

4.2 Stock Picking & Constraints

The stock picking based on the value and quality scores follows a set of constrains to refine the process, crucial for guiding our investment decisions. We are incline to a well-diversified portfolio and we do it by investing across all economic sectors available. Therefore, our industry constraint is applied across both long and short positions. It ensures a diversified portfolio across various industries by mandating the selection of an equal number of stocks from each industry available at the time of rebalancing, with a bias on the most populated sectors. This strategy is intended to reduce sector-specific risks and enhance the overall resilience of the portfolio against industry-specific downturns.

Some expedients for short-selling only have been undertaken given the challenges in systematically short selling companies. We implement a minimum market cap constraint, to ensure investments are made in relatively well-established and larger companies, with a threshold of $1 bil. This is aimed at mitigating the risks associated with volatile small-caps that might experience big price swings. Another constraint we implement on the short leg is to avoid exposure to companies that have already experienced a significant decline in value, specifically those with more than a 50% cumulative loss over the past 12 months. This approach is designed to avoid mispriced companies that might be subject of price rebounds after a significant correction, thereby protecting the portfolio from short positions moving against the intended direction.

Together, these constraints provide a balanced framework for stock selection, aiming to optimize the risk-reward ratio and ensure a stable and diversified L/S portfolio.

4.3 Portfolio Construction

4.3.1 Weighting Methods

In our long-short equity strategy, we employ a variety of portfolio weighting methods to optimize performance and manage risk. These methods include:

1. Equally Weighted: This approach assigns an equal weight to each asset in the portfolio, irrespective of its market value or volatility. It promotes diversification by ensuring no single asset disproportionately influences the portfolio's performance.

2. Risk-Parity (Equal Risk Distribution): In this method, the portfolio is constructed in a way that each asset contributes equally to the overall risk. It aims to achieve a balanced risk distribution across all assets, rather than equal monetary allocation.

3. Target Beta: This strategy involves adjusting the portfolio weights to achieve a target beta equal to 1 in the long and short legs, respectively. The objective is to create a market-neutral long-short portfolio (beta of 0). This method is only relevant for symmetric long-short portfolios (e.g. 100/100).

Each of these methods has unique advantages and considerations, which are integrated into our back-testing process to evaluate their efficacy in various market conditions and scenarios. Details about the weighting methods can be found in the Appendix section.

4.3.2 L/S Equity Strategy

After constructing the systematic stock picking, we explore various market exposure levels in the long-short settings to evaluate their impact on portfolio performance. These ratios indicate the extent of long positions relative to short positions. The 100/100 strategy aims for market neutrality, balancing long and short positions equally. The 130/30 and 120/50 strategies, being slightly more aggressive, involve taking on more long positions than short. The 300/200 configuration represents a highly leveraged approach, with a significantly greater long exposure. By back-testing across these diverse exposures, we aim to understand the risk-return profile under various configurations.

4.3.3 L/S Portfolio Accounting

In our back-testing methodology, we implement a long-short portfolio accounting system. For instance, with an initial cash balance of 100, we short sell stocks worth 30 and simultaneously take long positions worth 130. This approach keeps our Net Asset Value (NAV) at 100, maintaining a balanced yet leveraged portfolio. This strategy allows us to explore the effects of leveraging on portfolio returns and risk, while keeping the initial investment constant. Such a system is crucial for understanding the dynamics of a leveraged long-short equity strategy.

4.3.4 Turnover & Transactions Cost

We also incorporate transaction costs and borrowing expenses into our back-testing. We apply a transaction cost of 20 basis points (bps) on the portfolio's turnover and a borrowing cost on the short position value calculated as the risk-free rate plus a 50 bps spread. This inclusion of real-world financial costs is essential for a more accurate assessment of the strategy's potential performance, providing a clearer picture of net returns after accounting for these expenses.

4.4 Hyperparametrization

To enhance the performance and assess the robustness of our L/S equity strategy, we conduct a systematic grid search for parameters optimization. The grid search focuses on identifying the most effective combination of parameters under different configurations. Specifically, we explore variations in parameters related to stock-picking style (i.e., value, quality, value-quality), rebalancing frequencies, number of stocks invested for each leg, and the weighting methods for portfolio construction.

The performance of each portfolio is rigorously evaluated using key metrics such as Sharpe ratio, annualized return and volatility, maximum drawdown, and average turnover. The grid search results in 5760 portfolios, providing insights into the sensitivity of our strategy to changes in specific parameters and facilitating the identification of parameter values that contribute to optimal portfolio performance.

The results of the grid search, including key performance metrics, are presented under each exposure level (100/100, 130/30, 120/50, and 300/200).

5 Results

5.1 Stock Picking Performance Assessment under Market Neutral Setting

In this section, we analyze whether our systematic stock selection enables us to effectively choose stocks for constructing a long-short portfolio. Our goal here is not to select the absolute best market-neutral portfolios. Instead, we aim to evaluate the performance of our systematic stock picking across various investment styles.

For each stock picking signal, Value, Quality, Value & Quality, Value & Quality & Momentum, we consider three weighting methods to construct the long and short positions: equal weighting (EW), beta targeting (MN), and risk parity (RP). The portfolios are rebalanced monthly. We implement an industry constraint on the selection of the different stocks. We consider both cases with no transaction cost (Table 3) and with a transaction cost of 20bps on the turnover (Table 4).

The initial observation reveals that in all the simulated portfolios, the NAV consistently remains above 0, with the minimum NAV always being positive. This suggests that there is no systematic error in our stock selection for the construction of long-short portfolios.

Furthermore, examining the maximum market beta achieved by any of these portfolios, calculated using the Fama-French 3 factors plus momentum model (Carhart 4 factors model), reveals a value of 0.023 (refer to Table 3). This implies that none of these market-neutral (100/100) portfolios exhibit positive market exposure. However, some portfolios demonstrate a notable negative market exposure.

Even more remarkably, a majority of these portfolios have produced a significantly positive alpha (refer to Table 3). Taking transaction costs into account (see Table 4), it becomes evident that the portfolio yielding the most significant alpha is constructed with the Value & Quality signal. This underscores the notion that incorporating quality in addition to value serves to improve the performance of a value-only portfolio.

5.2 Practical Implementation under Different Market Exposures

Building on the insights gained from the grid search and our predefined exposure, we present the practical implementation of our L/S equity strategy under varying market exposures: 100/100, 130/30, 120/50, and 300/200. The grid search allowed us to identify optimal parameters combinations for each exposure setting, refining our strategy for enhanced risk-adjusted performance, as measured in terms of Sharpe ratio. For each configuration, we present returns assuming 20bps transactions cost on the turnover. Moreover, we analyse the exposure to Fama-French 3-factors plus momentum, and finally we analyze the impact on performance of adding borrowing cost to the short selling.

5.2.1 100/100: Market Neutral

In this market-neutral configuration, noteworthy results are achieved with the target beta method (MN), removing the industry constraint (NI), and rebalancing quarterly (Q). The value-quality signal on both the long and short legs yields a Sharpe ratio of 0.591. This Sharpe ratio is further increased to 0.627 by incorporating momentum into the long stock-picking, while keeping value-quality as the short signal (refer to Table 5).

When running the portfolio with value (VAL) and quality (QLT) separately, the optimal parameters indicate an equal weighting method with yearly rebalancing. A Sharpe ratio of 0.461 is achieved in the value portfolio with an industry constraint, whereas quality performs better without the industry constraint, yielding a Sharpe ratio of 0.343. These two portfolios are significantly impacted by adding borrowing costs to the short leg (see Table 7). In terms of Sharpe ratio, the value portfolio loses 34% of its performance, while quality experiences an even greater impact, losing about 43% of its Sharpe ratio. The more resilient portfolios, when adding this extra layer of costs, are the value-quality portfolio (losing only 14%) and the combination of value-quality-momentum on the long and value-quality on the short (losing only 13%). This provides strong indications that integrating quality into value offers a more robust solution than relying on value alone.

Moreover, the factor exposures metrics provide valuable insights into the strategy's exposure with specific market factors (Table 6). Notable factor exposures include a strong positive tilt towards value stocks, as indicated by high exposures to the High Minus Low (HML) factor in various portfolios. The strategies also exhibit partial hedges against overall market movements, with negative exposures to the market factor (MKTRF) serving as a risk mitigation measure. Positive alphas across different factors suggest the strategy's ability to generate excess returns beyond what can be attributed to the FF3-MoM model. Additionally, the factor exposures underscore the strategy's focus on risk factors such as size (SMB) and momentum (UMD). These findings highlights the underlying characteristics of the 100/100 configuration, providing a foundation for further analysis and comparison with alternative exposure settings.

5.2.2 130/30: Alpha Extension

In Table 9, we explore various signal combinations to pinpoint the most promising strategy within this exposure profile. The top-performing portfolio, boasting a Sharpe ratio of 0.898, adopts equal weighting and monthly rebalancing. It involves taking long positions in value-quality stocks while simultaneously shorting poor-quality (junk) stocks.

Among the diverse 130/30 portfolios created, a recurring trend of positive and substantial alpha is evident (refer to Table 10). Notably, the exposure to the market factor remains significant but is consistently below 1. With the exception of the quality-only portfolio, all other portfolios exhibit substantial exposure to the value factor (HML) by design. Moreover, portfolios incorporating quality demonstrate lower R-squared values, suggesting an advantage over traditional models. Despite this, overall R-squared values remain high, emphasizing the effectiveness of the FF3-MoM model in explaining the excess returns.

Finally, the significant impact of transactions and borrowing costs on portfolio performances is noteworthy, especially pronounced in the monthly rebalanced portfolios, inherently featuring higher average turnover (refer to Table 11). Interestingly, the consistently superior performance of the portfolio adopting a long position in value-quality stocks and shorting poor quality stocks persists across various scenarios. This observation underscores the resilience and efficacy of this specific strategy despite cost fluctuations, emphasizing its robustness within the 130/30 configuration.

5.2.3 120/50: Mid-Range Strategy

Within the 120/50 exposure profile, the most compelling portfolio stands out with a notable Sharpe ratio of 0.961, revealing a strategic composition favoring value-quality stocks in the long position and (poor) quality stocks on the short side. Employing equal weighting and an industry constraint alongside monthly rebalancing, this portfolio's consistent out-performance underlines the significance of astute stock selection, showcasing its ability to thrive despite heightened turnover (Table 13).

Examining the factors exposure, all portfolios demonstrate positive and significant alpha (see Table 14). Remarkably, the exposure to the market factor remains substantial yet consistently below 1. Contrasted with the 130/30 setting, notably lower R-squared values are observed overall, particularly evident in the quality-only portfolio with an R-squared at 0.214. This suggests that the FF3-MoM model explains only a fraction of the excess performance, emphasizing the need for nuanced strategies beyond conventional factor models.

Even with the inclusion of transactions and borrowing costs (refer to Table 15), the best Sharpe ratio without costs of 1.119 decreases to 0.872, thus maintaining a noteworthy result. This observation underscores the potential for significant enhancements through the systematic implementation of a value-quality strategy, also emphasizing the importance of optimizing trading pipelines to minimize transaction costs.

5.2.4 300/200: High-Leverage Strategy

In the analysis of our most leveraged profile, a consistent observation emerges regarding the optimal performer, achieving an impressive Sharpe ratio of 1.053 (refer to Table 17). Once again, this portfolio strategically positions itself by taking long positions in value-quality stocks while shorting poor-quality stocks. The recurrence of this observation across diverse exposure profiles underscores the robustness and reliability of this particular long-short signal combination. Notably, the heightened leverage necessitates a significantly increased turnover, expressed as a percentage of Net Asset Value (NAV). Despite this elevated turnover, the systematic value-quality strategy yields a respectable Sharpe ratio, even when accounting for transaction costs, highlighting the efficiency of the meticulous stock-picking process. The calibrated ad-hoc procedure in selecting stocks for both long and short legs contributes significantly to this systematic strategy, validating the efficacy of our implementation process tailored to the hedge fund manager's exposure profile.

As observed across the other factor exposure profiles, there's a consistent generation of positive and significant alpha (Table 18). The exposure to the market factor, significant and consistently below 1 for most portfolios, except the quality-only portfolio, underscores their relatively limited correlation with overall market movements. Notably, the quality-only portfolio displays non-significant exposure to the market while revealing negative and highly significant exposure to size. This suggests that quality is long in large-cap stocks and short in small-cap stocks, within this specific portfolio.

The R-squared values observed within the 300/200 setting are notably the lowest encountered thus far. This emphasizes the limitation of the FF3 plus momentum model in explaining the excess performance within these leveraged portfolios, especially those integrating the quality factor. Moreover, the examination of transaction costs unveils a substantial impact on performance, witnessing the most significant decrease in Sharpe ratio among the various exposure profiles (Table 19). Specifically, the inclusion of transaction costs markedly diminishes the Sharpe ratio of the best portfolio by over 0.35. These costs, amplified due to increased turnover, highlight the critical need for strategic cost management in optimizing performance within highly leveraged exposure scenarios.

5.3 Performance Decomposition

For the best portfolio in each configuration (100/100, 130/30, 120/50, 300/200), we have decomposed the performance of the long and short positions. We also compare the performance of the long short portfolio to the long net market exposure adjusted portfolio (see Tables 21, 22, 23, 24, and Figures 2, 3, 4, 5).

We defined the long net market exposure (NME) adjusted portfolio return as the return of the long portfolio multiply by the difference in the long leg market exposure percentage minus the short leg market exposure percentage. What is remaining is invested at the risk-free rate. (Long NME Adjusted return = Long return $\times$ (% long $-$ % short) + rf $\times$ (1 $-$ (% long $-$ % short)).

Concerning the analysis of the long and short position performance in every configuration, the long portfolio has a much higher annualized mean return and Sharpe ratio than the short position. This consistent overperformance of the long portfolio highlights that the stock selection process in the long short portfolio construction is well-executed.

Moreover, concerning the performance of the long-short portfolio and the long net market exposure adjusted portfolio, the Sharpe ratio of the long-short portfolios is always significantly higher than the long net market-adjusted portfolio in all the different configurations. These results highlight the importance of the short position in the long short portfolio.

6 Sensitivity Analysis

We perform a sensitivity analysis with the aim to rigorously assess the robustness of the recurring pattern underscored in the previous findings. This analysis underpins the dominance of the long value-quality and short quality signals combination across diverse exposure profiles, notably in terms of risk-adjusted performance, as measured by the Sharpe ratio.

We proceed by developing and implementing an ad-hoc procedure to assess the validity of our results, both on positive and negative outcomes (contraposite). In particular, our process encompasses exploring all potential parameter combinations: from varying weighting methods for the long and short legs to different rebalancing frequencies, implementation of industry constraint, and selecting signals for both positions. We backtest a grand total of 5760 portfolios, namely 1152 portfolios for each of the 5 exposure profiles originally considered, evaluating their balance sheets and performance metrics of interest. The sensitivity analysis consists in two phases (each performed for the different exposure profiles):

- Success assessment: The analysis focuses on the top 200 portfolios with the highest Sharpe ratios, for the considered exposure profile, approximating 17.4% of the total (1152). The objective is to ascertain the persistence of the identified signal pattern—specifically, the prevalence of long value-quality and short quality signal combination—consistent with the primary findings.
- Default assessment: Scrutinizing defaulted portfolios within each exposure profile elucidates the predominant signals leading to defaults. Insights into factors contributing to defaults across diverse portfolio settings are also provided.

Interestingly, within the classic 130/30 exposure profile, no defaults are recorded (see Table 12). Among the top 200 portfolios with the highest Sharpe ratios, a substantial majority mirror the identified signal pattern: 43.5% long value-quality and 37.5% short quality. This reaffirms the prevalence of the pattern among successful portfolios, corroborating earlier findings.

Defaults exhibit a direct correlation with leverage, quantified by the Debt/Equity ratio. Profiles with higher leverage, such as the 300/200 profile, exhibit the most defaults at 22.2% (see Table 20), while the 130/30 profile demonstrates the lowest at 0%.

Analyzing across configurations reveals intriguing insights. The short signal predominantly associated with defaults is short value-quality-momentum, except within the 100/100 configuration where short value-quality performs poorly. Similarly, long value consistently leads to higher defaults, except within the 100/100 portfolio where it is either long value-quality or long value-quality-momentum.

In conclusion, the resilience of the identified signal pattern — specifically, taking long positions in value-quality stocks and short positions in poor-quality stocks — is consistently validated across diverse portfolio settings. This underscores the effectiveness of incorporating quality into our client's systematic strategy, particularly within the long leg trades. However, it is crucial to mitigate risks by refining the stock-picking process with various constraints. Many defaults encountered were typically associated with highly leveraged portfolios using the target beta method, annual rebalancing, and lacking an industry constraint. These issues could be prevented with improved risk management and position sizing, especially in strategies without industry allocation, which can lead to dangerously concentrated short positions.

Examining the overall portfolio defaults, where 352 out of 5760 portfolios defaulted, the common characteristics of the defaulted portfolios precisely align with the characteristics just explained. In Table 25, we break down the total portfolio defaults to assess the weaknesses of our strategies. Notably, the target beta with yearly rebalancing emerges as the least favorable combination, as the original intuition behind this strategy is designed with a monthly rebalancing frequency.

7 Conclusion

This study showcases the efficacy of a systematic long-short equity strategy integrating quality and value factors. Backed by robust academic insights and meticulous data analysis, our strategy consistently demonstrates resilience and potential for market out-performance across diverse market exposures. The findings validate the significance of quality at a good price, offering a compelling investment avenue for value-driven portfolios in the US equity market.

Drawing on sound literature, we developed a systematic approach targeting high value-quality long positions while identifying short opportunities in stocks demonstrating poor quality (junk). The methodology, limited to a maximum of 25 stocks each on the long and short sides, allowed for concentrated positions.

Our results shine a light on the emergence of the long value-quality and short (poor) quality signal combination as the most compelling strategy. This signal pattern consistently outperforms traditional value-only strategies, showcasing increased risk-adjusted returns. Sensitivity analyses across various market exposures affirm the resilience and superiority of this integrated approach. Notably, this strategy maintains its edge even when subjected to transaction costs and market fluctuations, providing investors with a robust and reliable investment framework.

The sensitivity analysis conducted underlines the stability and durability of our findings, demonstrating the strategy's consistent out-performance under diverse scenarios. These results serve as a beacon for investors, highlighting the potential for enhanced risk-adjusted returns by adopting an integrated value-quality approach. Considerations for investors include the validation of this approach in navigating market uncertainties and its potential to deliver superior risk-adjusted returns.

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A Appendix

Weighting Methods

Risk Parity
Minimize the variance of risk contribution:

$$w^* = \arg\min\ f(w)$$
$$u.c = \begin{cases} 1'w = 1 \\ 0 \leq w_i \leq 1 \end{cases}$$

$$f(w) = \sum_{i=1}^{n} \sum_{j=1}^{n} (w_i(\Sigma w)_i - w_j(\Sigma w)_j)^2$$

w : vector of weights
Σ: covariance matrix

Target Beta
Set the portfolio beta to 1:

$$w^* = \arg\min\ f(w)$$
$$u.c = \begin{cases} 1'w = 1 \\ 0 \leq w_i \leq 1 \end{cases}$$

$$f(w) = (w'\beta - 1)^2$$

w : vector of weights
Σ: vector of Beta

Turnover of the L/S portfolio

L/S Portfolio TO = long pct $*$ Long TO + short pct $*$ Short TO

B Tables

Variable	Definition
Fundamental Quarterly:	
PERMNO	Historical CRSP PERMNO Link to COMPUSTAT Record (lpermno)
FQTR	Fiscal Quarter
CONM	Company Name
TIC	Ticker Symbol
EXCHG	Stock Exchange Code
GSECTOR	GIC Sectors
LOC	Current ISO Country Code - Headquarters
ATQ	Assets - Total
COGSQ	Cost of Goods Sold
CSHOQ	Common Shares Outstanding
DLCQ	Debt in Current Liabilities
DLTTQ	Long-Term Debt - Total
DPQ	Depreciation and Amortization - Total
LTQ	Liabilities - Total
NIQ	Net Income (Loss)
PIQ	Pretax Income
REQ	Retained Earnings
REVTQ	Revenue - Total
WCAPQ	Working Capital (Balance Sheet)
XINTQ	Interest and Related Expense- Total
CAPXY	Capital Expenditures
Security Monthly:	
PRCCM	Price - Close - Monthly
TRT1M	Monthly Total Return
Stock Monthly:	
BID	Bid Price
ASK	Ask Price
VOL	Volume
SPRTRN	Return on S&P Composite Index
Fama-French 3 Factors:	
MKTRF	Excess Return on the Market
SMB	Small-Minus-Big Return
HML	High-Minus-Low Return
UMD	Momentum Factor
RF	Risk-Free Return Rate (One Month Treasury Bill Rate)
Other Variables:	
QTR	Calendar Quarter
MTH	Calendar Month
PRCCQ	Price - Close - Quarterly
SPRDPCT	$(ASK - BID)/ASK$
DVOL	$PRCCM \times VOL$
WCAPCHQ	Working Capital Changes - Total - Quarterly
CAPXQ	Capital Expenditures Quarterly

Table 1: Variable Definitions

Variable	Definition
BE	Total Assets – Total Liabilities: $ATQ - LTQ$
ME	Price times Total Shares Outstanding: $PRCCQ * CSHOQ$
BE/ME	Book to Market Equity ratio: BE/ME
E/P	Earnings to Price ratio: $(NIQ/CSHOQ)/PRCCM$
CF/P	Cash Flow to Price ratio: Net Income plus Depreciation minus Changes in Working Capital and Capital Expenditures per Share Divided by Price: $(NIQ + DPQ - WCAPCHQ - CAPXQ)/CSHOQ/PRCCM$
GPOA	Gross Profits over Assets: Revenue minus Costs of Goods Sold divided by Total Assets: $(REVTQ - COGSQ)/ATQ$
ROE	Return on Equity : Net Income divided by Book-Equity: NIQ/BE
ROA	Return on Asset : Net Income divided by Total Assets: NIQ/ATQ
CFOA	Cash Flow over Assets: Net Income plus Depreciation minus Changes in Working Capital and Capital Expenditures divided by Total Assets: $(NIQ + DPQ - WCAPCHQ - CAPXQ)/ATQ$
GMAR	Gross Margin: Revenue minus Costs of Goods Sold divided by Total Sales: $(REVTQ - COGSQ)/REVTQ$
ACC	Accruals: Depreciation minus Changes in Working Capital $-(WCAPCHQ - DPQ)/ATQ$
Leverage	Sum of Long-Term Debt, Short-Term Debt over Total Assets: $-(DLTTQ + DLCQ)/ATQ$
Altman's Z-Score	Measure the company's likelihood of bankruptcy, $(1.2WC + 1.4RE + 3.3EBIT + 0.6ME + SALE)/AT$: $(1.2WCAPQ + 1.4REQ + 3.3(PIQ + XINTQ) + 0.6ME + REVTQ)/ATQ$
Beta	Beta over 5 years (monthly return), S&P 500 as the market portfolio

For the variables COGSQ, DPQ, NIQ, PIQ, REQ, REVTQ, WCAPCHQ, XINTQ, and CAPXQ we compute the last twelve months values, as standard industry practice to minimize noise in metrics due to business cyclicality and other accounting effects.

Table 2: Accounting Variable Definitions for Score Building

Weighting	SIG	MEAN	VOL	SHARPE	MDD	MDD_PRD	AVG_TO	α	t_α	β	t_β	$\min_{NAV}$
EW (100/100)	VAL	0.107	0.174	0.389	0.422	99-09_00-02	1.113	0.100	3.431	-0.078	-1.241	96.981
	QLT	0.082	0.146	0.289	0.441	20-07_21-05	0.924	0.088	3.605	-0.307	-6.073	96.619
	VQ	0.098	0.148	0.393	0.527	16-02_21-06	1.080	0.105	3.959	-0.300	-5.403	95.985
	VQAM	0.140	0.132	0.763	0.195	20-04_20-12	2.754	0.114	4.536	-0.083	-1.474	99.174
MN (100/100)	VAL	0.108	0.183	0.375	0.355	19-01_20-03	1.346	0.091	2.607	-0.037	-0.498	97.010
	QLT	0.078	0.157	0.246	0.346	05-10_08-08	1.120	0.059	1.849	-0.039	-0.625	98.458
	VQ	0.082	0.162	0.263	0.502	10-12_20-10	1.270	0.065	1.991	-0.095	-1.394	96.972
	VQAM	0.149	0.157	0.698	0.331	05-10_07-11	2.956	0.122	3.874	0.023	0.341	97.753
RP (100/100)	VAL	0.098	0.150	0.389	0.303	19-01_20-03	1.176	0.077	3.172	0.019	0.364	97.679
	QLT	0.078	0.130	0.296	0.381	20-07_21-02	0.960	0.074	3.251	-0.207	-4.086	98.196
	VQ	0.085	0.125	0.365	0.435	16-02_21-04	1.138	0.078	3.371	-0.184	-3.703	97.500
	VQAM	0.129	0.120	0.746	0.190	20-04_20-12	2.802	0.098	4.289	-0.036	-0.655	98.992

Portfolios Parameters: 25 long stocks, 25 short stocks, Industry Constraint = Yes,
Rebalancing Frequency = Monthly, Transactions Cost = 0bps on TO

Table 3: Market Neutral (100/100) Portfolio no Transactions Cost

Weighting	SIG	MEAN	VOL	SHARPE	MDD	MDD_PRD	AVG_TO	α	t_α	β	t_β	$\min_{NAV}$
EW (100/100)	VAL	0.080	0.174	0.234	0.431	99-09_00-02	1.113	0.072	2.502	-0.080	-1.281	94.103
	QLT	0.060	0.147	0.136	0.451	20-07_21-05	0.924	0.065	2.680	-0.309	-6.142	96.162
	VQ	0.072	0.148	0.217	0.587	16-02_21-06	1.080	0.079	2.982	-0.304	-5.506	95.456
	VQAM	0.073	0.132	0.257	0.359	15-12_20-12	2.754	0.048	1.915	-0.089	-1.604	98.102
MN (100/100)	VAL	0.076	0.183	0.197	0.425	10-04_15-01	1.346	0.058	1.667	-0.039	-0.523	95.896
	QLT	0.051	0.157	0.073	0.400	05-10_08-08	1.120	0.031	0.992	-0.043	-0.680	97.861
	VQ	0.051	0.162	0.073	0.626	10-12_20-10	1.270	0.034	1.049	-0.099	-1.459	96.375
	VQAM	0.078	0.156	0.243	0.432	05-03_07-11	2.956	0.050	1.610	0.016	0.243	96.092
RP (100/100)	VAL	0.069	0.149	0.199	0.325	19-01_20-03	1.176	0.048	1.994	0.016	0.319	97.031
	QLT	0.055	0.130	0.118	0.389	20-07_21-02	0.960	0.051	2.232	-0.209	-4.155	97.672
	VQ	0.058	0.125	0.143	0.510	16-02_21-04	1.138	0.051	2.194	-0.188	-3.826	96.908
	VQAM	0.061	0.119	0.180	0.350	06-09_10-07	2.802	0.030	1.335	-0.042	-0.783	97.601

Portfolios Parameters: 25 long stocks, 25 short stocks, Industry Constraint = Yes,
Rebalancing Frequency = Monthly, Transactions Cost = 20bps on TO

Table 4: Market Neutral (100/100) Portfolio with Transactions Cost

NAME (100/100)	SHARPE	MEAN	VOL	MDD	MDD_PRD	CALMAR	AVG_TO
VAL_25_EW_VAL_20_EW_I_Y	0.461	0.111	0.156	0.260	16-04_19-08	0.277	0.280
QLT_25_EW_QLT_20_EW_NI_Y	0.343	0.101	0.180	0.479	98-08_00-08	0.129	0.255
VQ_25_MN_VQ_20_MN_NI_Q	0.591	0.216	0.298	0.604	17-05_19-08	0.292	0.795
VQAM_25_MN_VQAM_15_MN_I_Q	0.358	0.106	0.186	0.510	02-10_11-07	0.131	1.105
VQAM_20_MN_VQ_20_MN_NI_Q	0.627	0.229	0.302	0.560	17-05_19-08	0.338	0.972

Portfolios parameters: Long Signal, # Long Stocks, Long Weighting Method, Short Signal, # Short Stocks, Short Weighting Method, Industry Constrain (Yes= I, No=NI), Rebalancing Frequency

Table 5: Strategies Performance with 100/100 Configuration

NAME (100/100)	SHARPE	α	β_{MKTRF}	β_{SMB}	β_{HML}	β_{UMD}	R^2
VAL_25_EW_VAL_20_EW_I_Y	0.461	0.100	-0.061	-0.042	0.483	-0.284	0.285
		3.539	-0.964	-0.322	5.212	-2.949	
QLT_25_EW_QLT_20_EW_NI_Y	0.343	0.119	-0.380	-0.368	-0.207	0.032	0.206
		3.725	-6.280	-3.830	-2.064	0.466	
VQ_25_MN_VQ_20_MN_NI_Q	0.591	0.217	-0.248	-0.128	0.631	-0.257	0.122
		3.881	-2.127	-0.715	2.960	-1.805	
VQAM_25_MN_VQAM_15_MN_I_Q	0.358	0.078	-0.013	0.145	0.289	0.025	0.032
		2.045	-0.150	1.078	2.476	0.223	
VQAM_20_MN_VQ_20_MN_NI_Q	0.627	0.218	-0.195	-0.056	0.635	-0.125	0.087
		4.022	-1.693	-0.312	3.213	-1.055	

Table 6: Factor Exposure Analysis for 100/100 Configuration

NAME (100/100)	MEAN	VOL	SHARPE	MDD
VAL_25_EW_VAL_20_EW_I_Y	0.118	0.156	0.506	0.244
VAL_25_EW_VAL_20_EW_I_Y_TC	0.111	0.156	0.461	0.260
VAL_25_EW_VAL_20_EW_I_Y_TC_BC	0.088	0.159	0.304	0.291
QLT_25_EW_QLT_20_EW_NI_Y	0.107	0.179	0.378	0.471
QLT_25_EW_QLT_20_EW_NI_Y_TC	0.101	0.180	0.343	0.479
QLT_25_EW_QLT_20_EW_NI_Y_TC_BC	0.076	0.186	0.194	0.561
VQ_25_MN_VQ_20_MN_NI_Q	0.235	0.298	0.654	0.584
VQ_25_MN_VQ_20_MN_NI_Q_TC	0.216	0.298	0.591	0.604
VQ_25_MN_VQ_20_MN_NI_Q_TC_BC	0.190	0.300	0.503	0.626
VQAM_25_MN_VQAM_15_MN_I_Q	0.133	0.187	0.501	0.390
VQAM_25_MN_VQAM_15_MN_I_Q_TC	0.106	0.186	0.358	0.510
VQAM_25_MN_VQAM_15_MN_I_Q_TC_BC	0.080	0.186	0.220	0.626
VQAM_20_MN_VQ_20_MN_NI_Q	0.252	0.302	0.704	0.533
VQAM_20_MN_VQ_20_MN_NI_Q_TC	0.229	0.302	0.627	0.560
VQAM_20_MN_VQ_20_MN_NI_Q_TC_BC	0.204	0.303	0.541	0.582

Table 7: Transactions Cost Analysis for 100/100 Configuration

SIG (100/100)	TOP_COUNT	TOP_PCT	DEF_COUNT	DEF_PCT
L_VAL	89	0.445	18	0.237
L_QLT	10	0.050	18	0.237
L_VQ	50	0.250	20	0.263
L_VQAM	51	0.255	20	0.263
S_VAL	24	0.120	8	0.105
S_QLT	74	0.370	16	0.211
S_VQ	74	0.370	30	0.395
S_VQAM	28	0.140	22	0.289

Table 8: Sensitivity Analysis for 100/100 Configuration

NAME (130/30)	SHARPE	MEAN	VOL	MDD	MDD_PRD	CALMAR	AVG_TO
VAL_25_EW_VAL_20_EW_I_Y	0.690	0.204	0.238	0.649	07-05_09-02	0.253	0.223
QLT_20_EW_QLT_15_EW_I_M	0.752	0.168	0.171	0.387	07-09_09-02	0.332	0.747
VQ_25_EW_VQ_20_EW_I_M	0.809	0.197	0.195	0.409	07-05_09-02	0.386	0.909
VQAM_25_EW_VQAM_15_EW_I_M	0.716	0.183	0.200	0.576	07-05_09-02	0.249	2.317
VQ_25_EW_QLT_15_EW_I_M	0.898	0.210	0.190	0.394	07-05_09-02	0.432	0.915

Table 9: Strategies Performance with 130/30 Configuration

NAME (130/30)	SHARPE	α	β_{MKTRF}	β_{SMB}	β_{HML}	β_{UMD}	R^2
VAL_25_EW_VAL_20_EW_I_Y	0.690	0.103	0.915	0.371	0.699	-0.365	0.715
		3.630	14.651	3.062	7.998	-3.206	
QLT_20_EW_QLT_15_EW_I_M	0.752	0.075	0.763	0.021	0.118	0.076	0.471
		3.005	15.291	0.261	1.515	1.638	
VQ_25_EW_VQ_20_EW_I_M	0.809	0.103	0.800	0.238	0.527	-0.188	0.643
		4.170	15.028	3.062	6.413	-2.986	
VQAM_25_EW_VQAM_15_EW_I_M	0.716	0.065	0.971	0.203	0.420	0.045	0.638
		2.648	16.134	1.931	5.432	0.748	
VQ_25_EW_QLT_15_EW_I_M	0.898	0.117	0.786	0.209	0.475	-0.163	0.621
		4.747	15.297	2.701	5.926	-2.689	

Table 10: Factor Exposure Analysis for 130/30 Configuration

NAME (130/30)	MEAN	VOL	SHARPE	MDD
VAL_25_EW_VAL_20_EW_I_Y	0.209	0.238	0.711	0.645
VAL_25_EW_VAL_20_EW_I_Y_TC	0.204	0.238	0.690	0.649
VAL_25_EW_VAL_20_EW_I_Y_TC_BC	0.197	0.238	0.663	0.657
QLT_20_EW_QLT_15_EW_I_M	0.186	0.171	0.857	0.370
QLT_20_EW_QLT_15_EW_I_M_TC	0.168	0.171	0.752	0.387
QLT_20_EW_QLT_15_EW_I_M_TC_BC	0.160	0.171	0.706	0.393
VQ_25_EW_VQ_20_EW_I_M	0.220	0.196	0.920	0.386
VQ_25_EW_VQ_20_EW_I_M_TC	0.197	0.195	0.809	0.409
VQ_25_EW_VQ_20_EW_I_M_TC_BC	0.190	0.195	0.769	0.419
VQAM_25_EW_VQAM_15_EW_I_M	0.239	0.201	0.992	0.531
VQAM_25_EW_VQAM_15_EW_I_M_TC	0.183	0.200	0.716	0.576
VQAM_25_EW_VQAM_15_EW_I_M_TC_BC	0.175	0.200	0.677	0.583
VQ_25_EW_QLT_15_EW_I_M	0.232	0.190	1.013	0.368
VQ_25_EW_QLT_15_EW_I_M_TC	0.210	0.190	0.898	0.394
VQ_25_EW_QLT_15_EW_I_M_TC_BC	0.202	0.190	0.856	0.404

Table 11: Transactions Cost Analysis for 130/30 Configuration

SIG (130/30)	TOP_COUNT	TOP_PCT	DEF_COUNT	DEF_PCT
L_VAL	37	0.185	0	nan
L_QLT	14	0.070	0	nan
L_VQ	87	0.435	0	nan
L_VQAM	62	0.310	0	nan
S_VAL	32	0.160	0	nan
S_QLT	75	0.375	0	nan
S_VQ	53	0.265	0	nan
S_VQAM	40	0.200	0	nan

Table 12: Sensitivity Analysis for 130/30 Configuration

NAME (120/50)	SHARPE	MEAN	VOL	MDD	MDD_PRD	CALMAR	AVG_TO
VAL_25_EW_VAL_20_EW_I_Y	0.693	0.175	0.195	0.527	07-05_09-02	0.257	0.237
QLT_20_EW_QLT_15_EW_I_M	0.755	0.142	0.136	0.217	07-09_09-02	0.472	0.810
VQ_25_EW_VQ_15_EW_I_Y	0.809	0.154	0.141	0.312	07-05_09-02	0.366	0.241
VQAM_25_EW_VQAM_20_EW_I_M	0.679	0.147	0.159	0.438	07-05_09-03	0.246	2.413
VQ_25_EW_QLT_15_EW_I_M	0.961	0.181	0.147	0.270	19-02_20-03	0.522	0.964

Table 13: Strategies Performance with 120/50 Configuration

NAME (120/50)	SHARPE	α	β_{MKTRF}	β_{SMB}	β_{HML}	β_{UMD}	R^2
VAL_25_EW_VAL_20_EW_I_Y	0.693	0.102	0.615	0.248	0.627	-0.331	0.609
		3.717	10.141	2.025	7.269	-3.114	
QLT_20_EW_QLT_15_EW_I_M	0.755	0.079	0.431	-0.106	0.038	0.102	0.214
		3.300	9.044	-1.514	0.491	2.344	
VQ_25_EW_VQ_15_EW_I_Y	0.809	0.092	0.457	0.161	0.274	-0.123	0.415
		4.151	9.982	1.706	4.153	-2.005	
VQAM_25_EW_VQAM_20_EW_I_M	0.679	0.056	0.665	0.118	0.347	0.083	0.460
		2.366	11.601	1.327	4.582	1.434	
VQ_25_EW_QLT_15_EW_I_M	0.961	0.118	0.451	0.068	0.367	-0.119	0.385
		4.941	9.418	0.983	4.828	-2.058	

Table 14: Factor Exposure Analysis for 120/50 Configuration

NAME (120/50)	MEAN	VOL	SHARPE	MDD
VAL_25_EW_VAL_20_EW_I_Y	0.181	0.196	0.721	0.522
VAL_25_EW_VAL_20_EW_I_Y_TC	0.175	0.195	0.693	0.527
VAL_25_EW_VAL_20_EW_I_Y_TC_BC	0.164	0.196	0.635	0.545
QLT_20_EW_QLT_15_EW_I_M	0.162	0.136	0.900	0.191
QLT_20_EW_QLT_15_EW_I_M_TC	0.142	0.136	0.755	0.217
QLT_20_EW_QLT_15_EW_I_M_TC_BC	0.129	0.136	0.659	0.231
VQ_25_EW_VQ_15_EW_I_Y	0.160	0.142	0.848	0.305
VQ_25_EW_VQ_15_EW_I_Y_TC	0.154	0.141	0.809	0.312
VQ_25_EW_VQ_15_EW_I_Y_TC_BC	0.142	0.141	0.728	0.338
VQAM_25_EW_VQAM_20_EW_I_M	0.206	0.160	1.041	0.375
VQAM_25_EW_VQAM_20_EW_I_M_TC	0.147	0.159	0.679	0.438
VQAM_25_EW_VQAM_20_EW_I_M_TC_BC	0.134	0.159	0.597	0.454
VQ_25_EW_QLT_15_EW_I_M	0.204	0.147	1.119	0.255
VQ_25_EW_QLT_15_EW_I_M_TC	0.181	0.147	0.961	0.270
VQ_25_EW_QLT_15_EW_I_M_TC_BC	0.167	0.147	0.872	0.280

Table 15: Transactions Cost Analysis for 120/50 Configuration

SIG (120/50)	TOP_COUNT	TOP_PCT	DEF_COUNT	DEF_PCT
L_VAL	48	0.240	7	0.350
L_QLT	14	0.070	2	0.100
L_VQ	76	0.380	6	0.300
L_VQAM	62	0.310	5	0.250
S_VAL	26	0.130	8	0.400
S_QLT	85	0.425	1	0.050
S_VQ	57	0.285	0	0.000
S_VQAM	32	0.160	11	0.550

Table 16: Sensitivity Analysis for 120/50 Configuration

NAME (300/200)	SHARPE	MEAN	VOL	MDD	MDD_PRD	CALMAR	AVG_TO
VAL_25_EW_VAL_20_EW_I_Y	0.730	0.417	0.517	0.798	07-05_09-02	0.473	0.698
QLT_20_EW_QLT_15_EW_I_M	0.770	0.311	0.352	0.693	20-07_21-02	0.391	2.430
VQ_25_EW_VQ_20_EW_I_Y	0.917	0.313	0.298	0.502	07-05_08-11	0.546	0.703
VQAM_25_EW_VQAM_20_EW_I_M	0.716	0.295	0.357	0.638	07-05_09-04	0.401	7.021
VQ_25_EW_QLT_20_EW_I_M	1.053	0.386	0.329	0.625	16-12_21-02	0.554	2.788

Table 17: Strategies Performance with 300/200 Configuration

NAME (300/200)	SHARPE	α	β_{MKTRF}	β_{SMB}	β_{HML}	β_{UMD}	R^2
VAL_25_EW_VAL_20_EW_I_Y	0.730	0.344	0.884	0.180	1.515	-1.165	0.448
		3.842	4.202	0.371	6.144	-2.680	
QLT_20_EW_QLT_15_EW_I_M	0.770	0.267	0.159	-0.693	-0.134	0.366	0.070
		3.986	1.185	-3.955	-0.594	3.039	
VQ_25_EW_VQ_20_EW_I_Y	0.917	0.260	0.383	0.036	0.549	-0.252	0.134
		4.597	3.183	0.135	3.320	-1.610	
VQAM_25_EW_VQAM_20_EW_I_M	0.716	0.169	0.850	0.008	0.720	0.349	0.153
		2.519	5.450	0.037	3.428	2.230	
VQ_25_EW_QLT_20_EW_I_M	1.053	0.343	0.252	-0.227	0.613	-0.204	0.097
		5.357	2.002	-1.303	3.145	-1.345	

Table 18: Factor Exposure Analysis for 300/200 Configuration

NAME (300/200)	MEAN	VOL	SHARPE	MDD
VAL_25_EW_VAL_20_EW_I_Y	0.432	0.517	0.759	0.793
VAL_25_EW_VAL_20_EW_I_Y_TC	0.417	0.517	0.730	0.798
VAL_25_EW_VAL_20_EW_I_Y_TC_BC	0.378	0.520	0.651	0.827
QLT_20_EW_QLT_15_EW_I_M	0.369	0.352	0.938	0.682
QLT_20_EW_QLT_15_EW_I_M_TC	0.311	0.352	0.770	0.693
QLT_20_EW_QLT_15_EW_I_M_TC_BC	0.259	0.352	0.621	0.696
VQ_25_EW_VQ_20_EW_I_Y	0.328	0.299	0.967	0.489
VQ_25_EW_VQ_20_EW_I_Y_TC	0.313	0.298	0.917	0.502
VQ_25_EW_VQ_20_EW_I_Y_TC_BC	0.271	0.301	0.767	0.592
VQAM_25_EW_VQAM_20_EW_I_M	0.466	0.360	1.184	0.510
VQAM_25_EW_VQAM_20_EW_I_M_TC	0.295	0.357	0.716	0.638
VQAM_25_EW_VQAM_20_EW_I_M_TC_BC	0.243	0.357	0.570	0.712
VQ_25_EW_QLT_20_EW_I_M	0.454	0.330	1.257	0.528
VQ_25_EW_QLT_20_EW_I_M_TC	0.386	0.329	1.053	0.625
VQ_25_EW_QLT_20_EW_I_M_TC_BC	0.334	0.329	0.894	0.677

Table 19: Transactions Cost Analysis for 300/200 Configuration

SIG (300/200)	TOP_COUNT	TOP_PCT	DEF_COUNT	DEF_PCT
L_VAL	54	0.270	76	0.297
L_QLT	17	0.085	45	0.176
L_VQ	70	0.350	68	0.266
L_VQAM	59	0.295	67	0.262
S_VAL	27	0.135	64	0.250
S_QLT	75	0.375	49	0.191
S_VQ	54	0.270	61	0.238
S_VQAM	44	0.220	82	0.320

Table 20: Sensitivity Analysis for 300/200 Configuration

	L/S vs Long NME Adjusted					Long vs Short			
	LS	LS_{TC}	$LS_{TC\&BC}$	LA	LA_{TC}	L	L_{TC}	S	S_{TC}
MEAN	0.252	0.229	0.204	0.038	0.038	0.162	0.149	-0.077	-0.087
VOL	0.302	0.302	0.303	0.025	0.025	0.243	0.243	0.352	0.351
SHARPE	0.704	0.627	0.541	-0.064	-0.064	0.505	0.450	-0.331	-0.360
MDD	0.533	0.560	0.582	0.045	0.045	0.553	0.561	0.992	0.994
CALMAR	0.398	0.338	0.282	-0.036	-0.036	0.222	0.194	-0.117	-0.127
AVG_TO	0.972	0.972	0.972	0.000	0.000	0.559	0.559	0.413	0.413
NORM_HI	NaN	NaN	NaN	NaN	NaN	0.130	0.130	0.029	0.029

Table 21: Long and Short Performance Decomposition for 100/100 Best Configuration

	L/S vs Long NME Adjusted					Long vs Short			
	LS	LS_{TC}	$LS_{TC\&BC}$	LA	LA_{TC}	L	L_{TC}	S	S_{TC}
MEAN	0.232	0.210	0.202	0.189	0.175	0.189	0.175	0.046	0.033
VOL	0.190	0.190	0.190	0.189	0.188	0.189	0.188	0.256	0.256
SHARPE	1.013	0.898	0.856	0.793	0.719	0.793	0.719	0.025	-0.025
MDD	0.368	0.394	0.404	0.447	0.460	0.447	0.460	0.671	0.679
CALMAR	0.523	0.432	0.402	0.334	0.294	0.334	0.294	0.010	-0.009
AVG_TO	0.915	0.915	0.915	0.579	0.579	0.579	0.579	0.538	0.538
NORM_HI	NaN	NaN	NaN	NaN	NaN	0.000	0.000	0.001	0.001

Table 22: Long and Short Performance Decomposition for 130/30 Best Configuration

	L/S vs Long NME Adjusted					Long vs Short			
	LS	LS_{TC}	$LS_{TC\&BC}$	LA	LA_{TC}	L	L_{TC}	S	S_{TC}
MEAN	0.204	0.181	0.167	0.139	0.129	0.189	0.175	0.046	0.033
VOL	0.147	0.147	0.147	0.132	0.132	0.189	0.188	0.256	0.256
SHARPE	1.119	0.961	0.872	0.751	0.677	0.793	0.719	0.025	-0.025
MDD	0.255	0.270	0.280	0.324	0.335	0.447	0.460	0.671	0.679
CALMAR	0.645	0.522	0.457	0.306	0.267	0.334	0.294	0.010	-0.009
AVG_TO	0.964	0.964	0.964	0.406	0.406	0.579	0.579	0.538	0.538
NORM_HI	NaN	NaN	NaN	NaN	NaN	0.000	0.000	0.001	0.001

Table 23: Long and Short Performance Decomposition for 120/50 Best Configuration

	L/S vs Long NME Adjusted					Long vs Short			
	LS	LS_{TC}	$LS_{TC\&BC}$	LA	LA_{TC}	L	L_{TC}	S	S_{TC}
MEAN	0.454	0.386	0.334	0.189	0.175	0.189	0.175	0.057	0.044
VOL	0.330	0.329	0.329	0.189	0.188	0.189	0.188	0.247	0.247
SHARPE	1.257	1.053	0.894	0.793	0.719	0.793	0.719	0.070	0.019
MDD	0.528	0.625	0.677	0.447	0.460	0.447	0.460	0.641	0.650
CALMAR	0.785	0.554	0.435	0.334	0.294	0.334	0.294	0.027	0.007
AVG_TO	2.788	2.788	2.788	0.579	0.579	0.579	0.579	0.525	0.525
NORM_HI	NaN	NaN	NaN	NaN	NaN	0.000	0.000	0.001	0.001

Table 24: Long and Short Performance Decomposition for 300/200 Best Configuration

Weighting Methods	Rebalancing Frequency	Industry Constraint	L/S Exposure
EW = 11.9%	M = 19.9%	Yes = 23.3%	100/100 = 21.6%
RP = 8.2%	Q = 21.0%	No = 76.7%	120/50 = 5.7%
MN = 79.9%	Y = 59.1%		300/200 = 72.7%
100%	100%	100%	100%

Table 25: Total Portfolio Defaults Breakdown (Total Defaults = 352)

C Figures

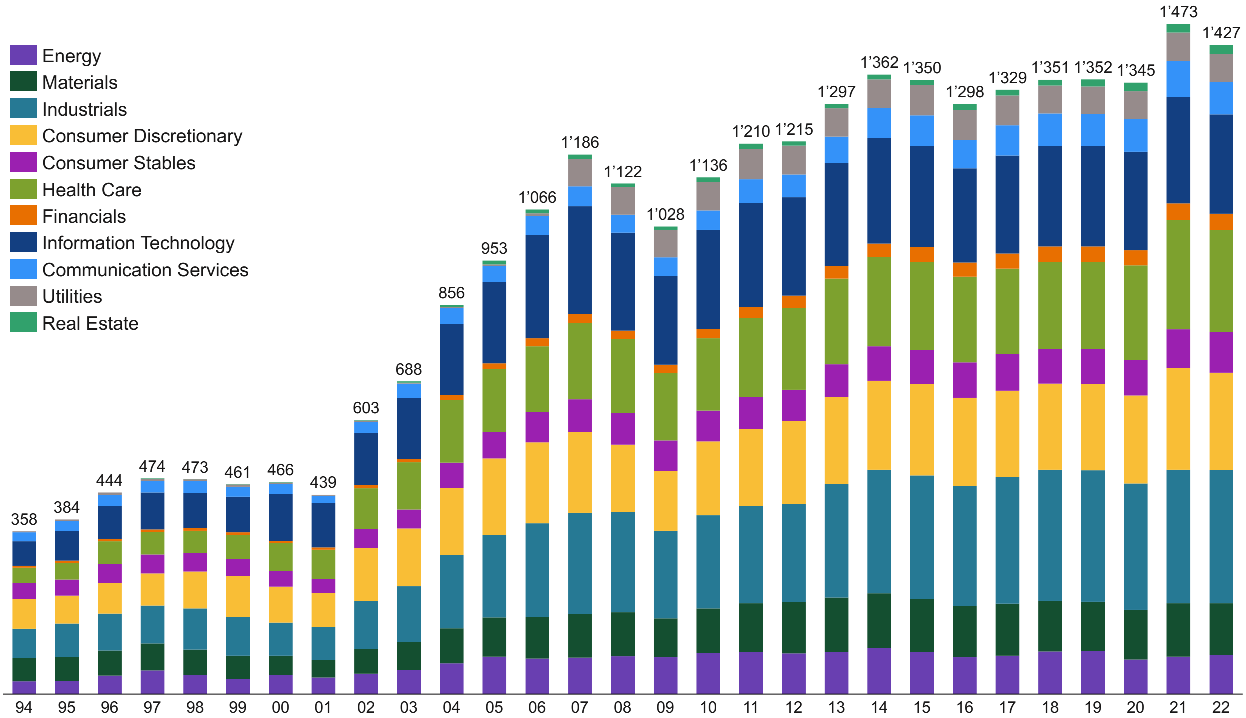

Figure 1: Final Stocks Universe

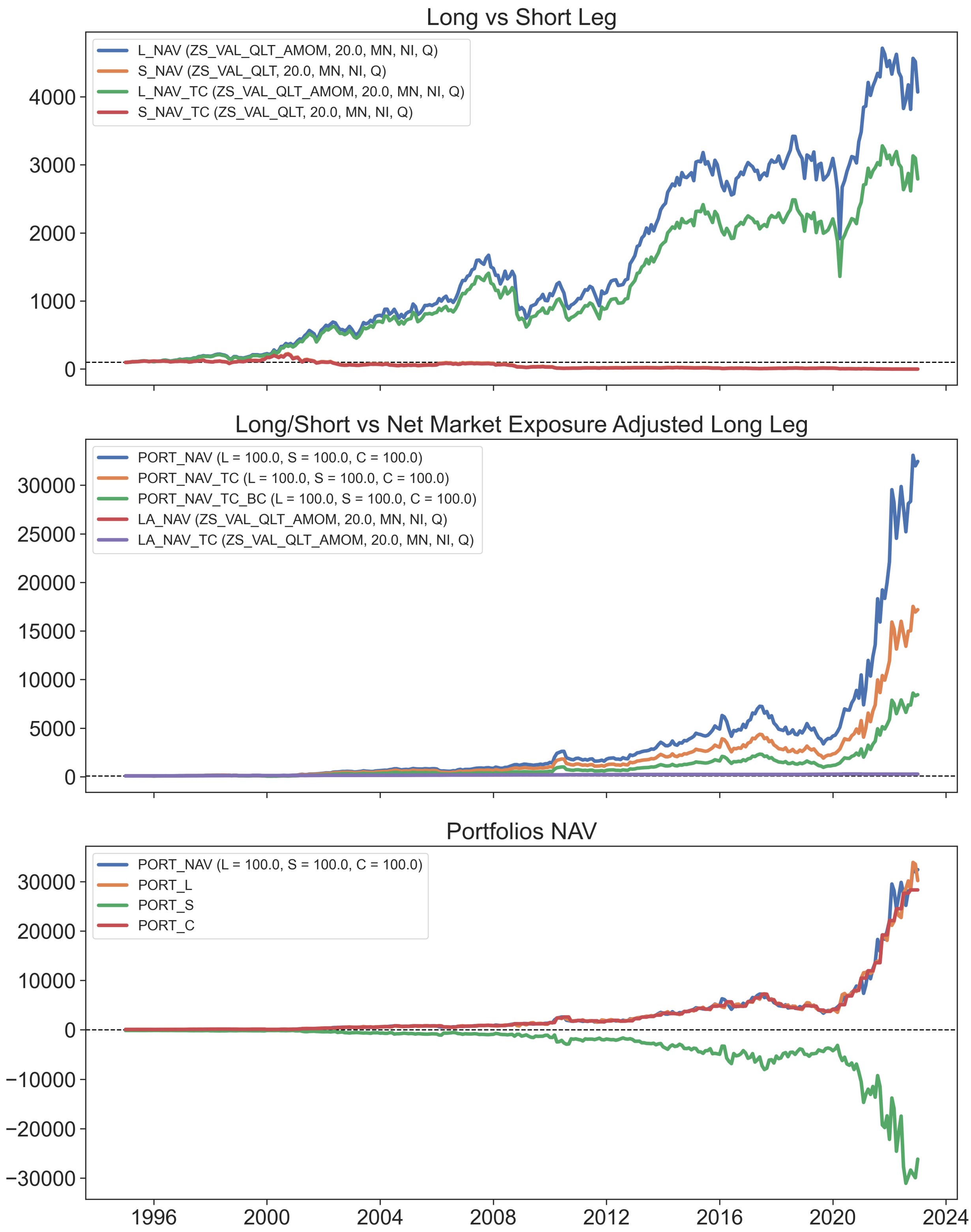

Figure 2: Long and Short Performance Decomposition for 100/100 Best Configuration

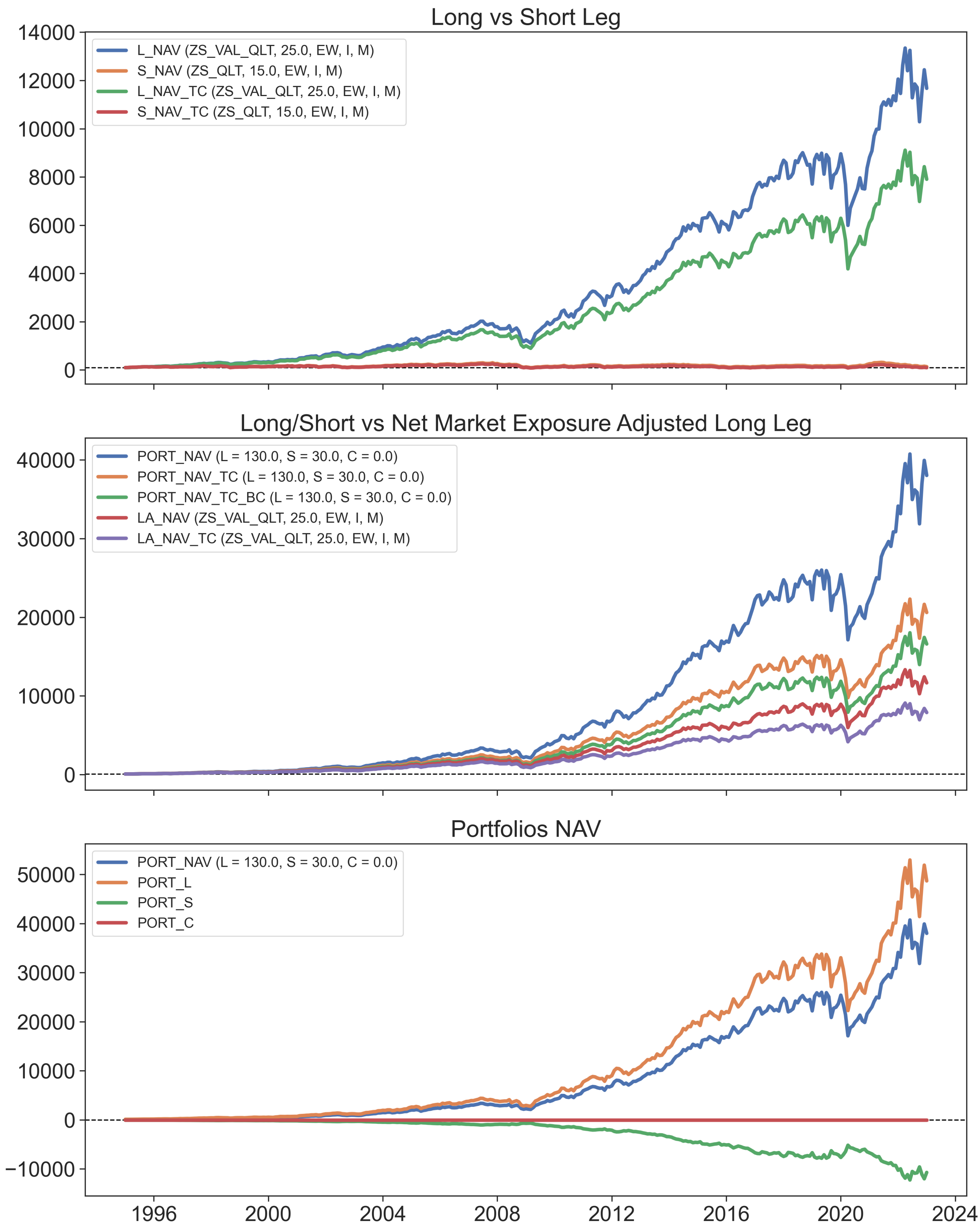

Figure 3: Long and Short Performance Decomposition for 130/30 Best Configuration

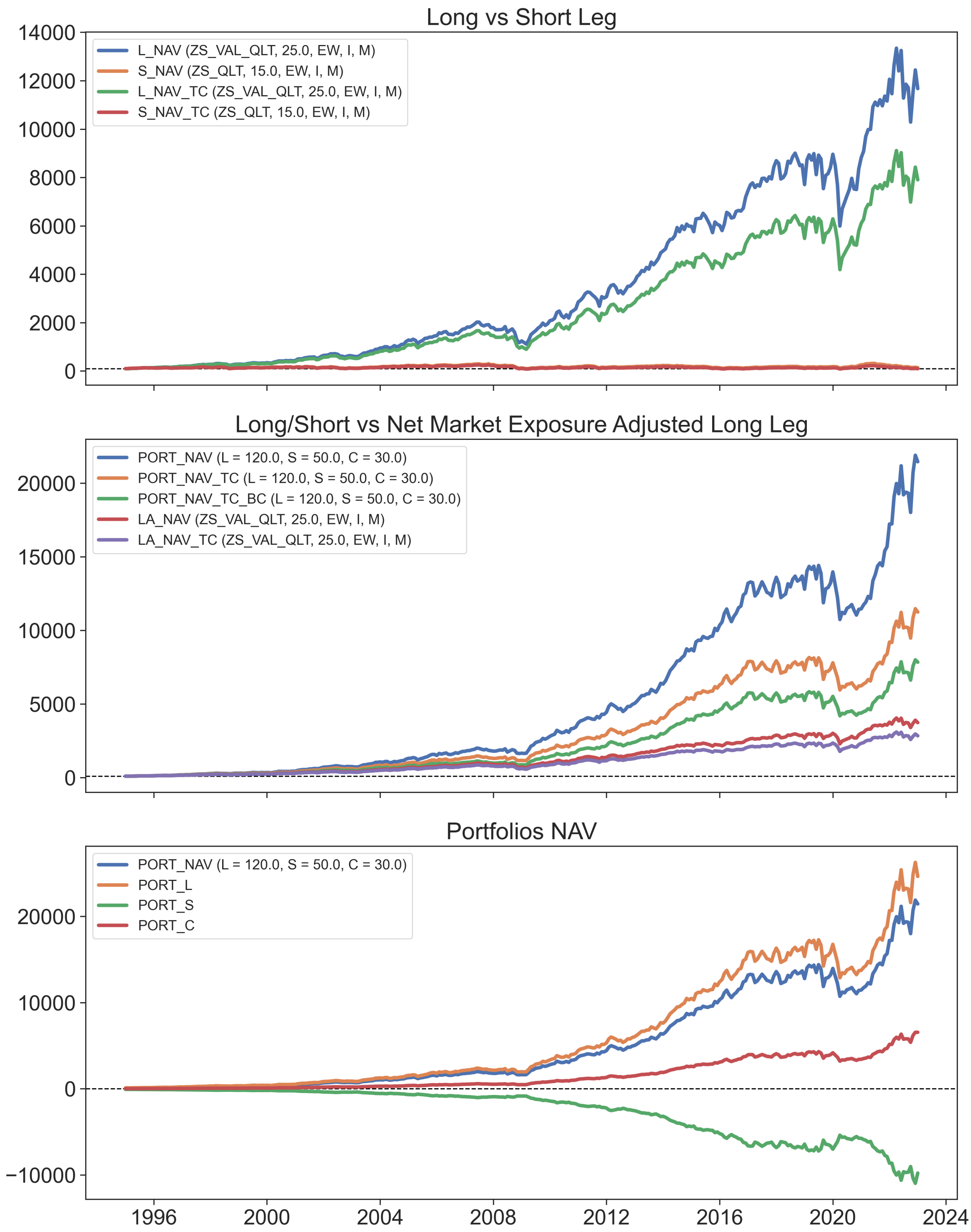

Figure 4: Long and Short Performance Decomposition for 120/50 Best Configuration

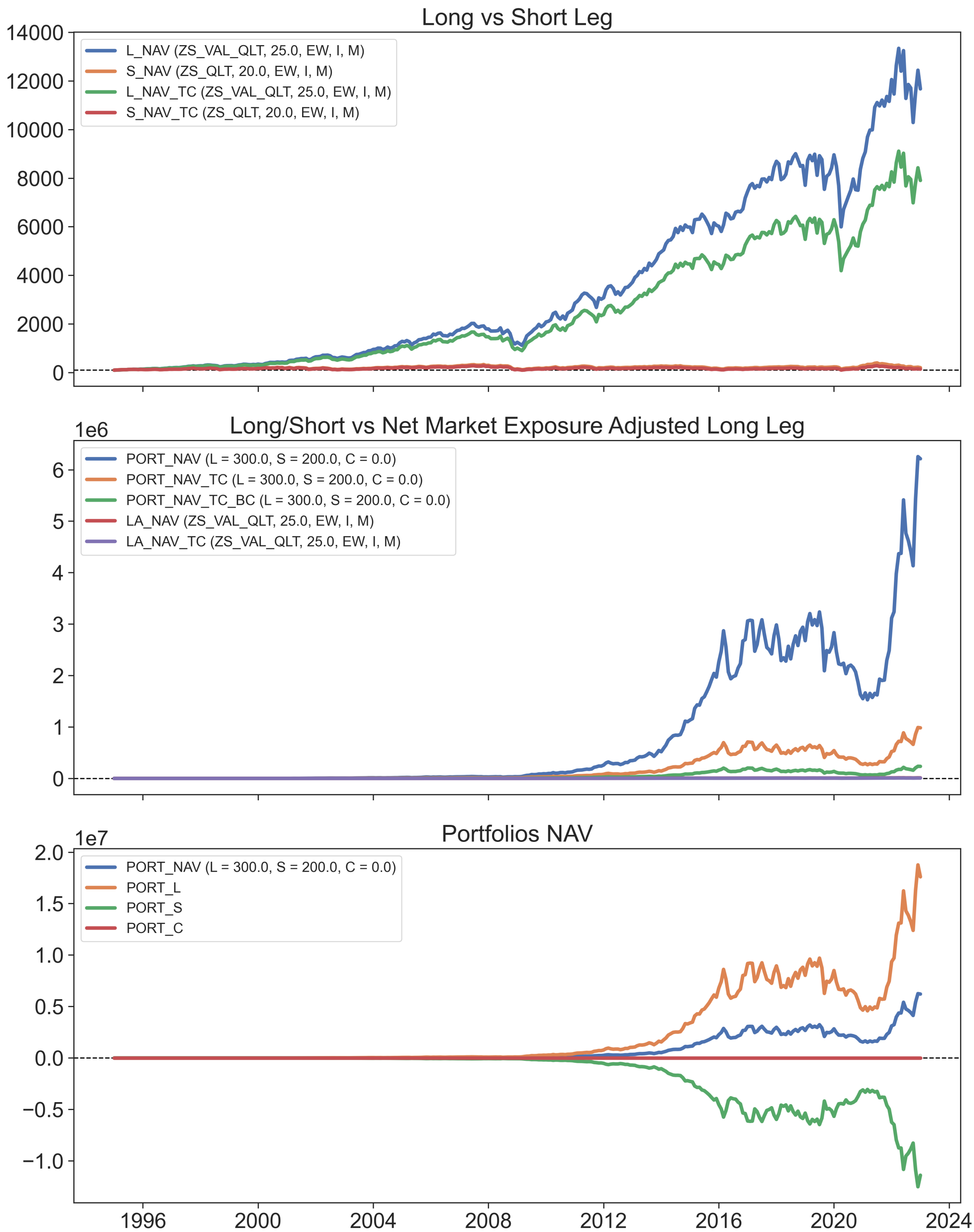

Figure 5: Long and Short Performance Decomposition for 300/200 Best Configuration